

LMS Topic 11

Autonomous Support & Lead Generation Agent

Project Documentation & Demonstration Summary

1. Project Overview

This project is an n8n-based autonomous customer-support and lead-generation workflow. It receives support tickets through a webhook, validates and normalizes the request, retrieves customer/account/ticket context from PostgreSQL, uses Gemini 2.5 Flash for structured AI analysis, routes the request based on the AI decision, sends a customer email, posts an internal Slack notification when required, logs token usage, and returns a structured webhook response.

2. Core Workflow

Stage	Purpose
01 – Support Webhook	Receives POST support-ticket requests.
02 – Validate Input	Checks required ticket/customer/request fields.
03 – Normalize Input	Creates a consistent internal ticket object.
04A/04B/04C – PostgreSQL	Retrieves customer profile, previous tickets, and account/lead data.
05 – Merge Context	Combines retrieved database branches.
06 – Prepare AI Context	Builds grounded context for the AI agent.
07 – AI Support Agent	Gemini analyzes category, priority, intent, lead status, and action.
08 – Parse & Validate AI	Validates the required structured JSON output.
09 – Decision Router	Routes to auto-resolve, lead follow-up, or human escalation.
10 – Dynamic DB Query	Retrieves relevant order/account data for downstream handling.
11 – Merge Routes	Converges routed branches for notification/email processing.
12 – Send Customer Email	Sends an automated customer response.
13 – Internal Notification	Posts support/lead information to Slack.
14 – Token Usage Logger	Stores AI token usage and execution information in PostgreSQL.
Build Success Response	Builds the final API response payload.
16 – Webhook Response	Returns the final JSON response to the caller.

3. AI Decision Logic

- Classifies requests into SUPPORT, BILLING, ACCOUNT, TECHNICAL, SALES, or GENERAL.
- Assigns LOW, MEDIUM, HIGH, or URGENT priority.
- Detects upgrade/purchase interest and marks the ticket as a lead.
- Uses auto-resolution only when the answer is low-risk and fully supported by retrieved context.
- Escalates ambiguous, high-risk, billing/payment, security, or uncertain cases to human review.
- Uses structured JSON output validation to keep downstream routing deterministic.

4. Database & Observability

The workflow uses PostgreSQL tables for customers, tickets, accounts, orders, and token_usage_log. The token logger records the ticket ID, customer ID, execution ID, model, input/output/total token counts, status, and timestamp when usage metadata is available. The workflow avoids fabricating token numbers.

5. Demonstration / Test Evidence

The final demonstration was tested with ticket TCK-5005. The test request completed successfully: the n8n workflow executed through the green/success states, the Slack internal notification was received in #support, and the customer email was received. The successful response returned an escalated/human-review result for the tested scenario.

Loom demonstration: <https://www.loom.com/share/650a95d3571a40d881b3c4c56249b80d>

6. Recommended GitHub Repository

Repository name: n8n-autonomous-support-lead-generation-agent

Recommended repository contents: exported n8n workflow JSON, README.md, sample test payloads, database setup SQL, screenshots/evidence, and a short architecture explanation. Never commit API keys, OAuth secrets, database passwords, or other credentials.

7. LMS Submission Description

Built a production-style Autonomous Support & Lead Generation Agent in n8n. The workflow accepts support tickets through a webhook, validates and normalizes input, retrieves customer history and account context from PostgreSQL, uses Gemini 2.5 Flash with structured JSON output to classify priority, intent, lead status and recommended action, routes tickets for auto-resolution, lead follow-up or human escalation, sends automated customer email responses, posts internal Slack notifications, logs AI token usage, and returns a structured webhook response. The final test with TCK-5005 was successfully demonstrated with workflow execution, Slack notification, and email delivery.

8. Skills Demonstrated

n8n workflow automation • Webhooks & API integration • PostgreSQL • Gemini / AI agent integration • Structured JSON output validation • Decision routing & escalation • Gmail/email automation • Slack internal notifications • Token usage logging • Error/fallback handling

Source workflow: LMS Topic 11 – Autonomous Support & Lead Generation Agent. Prepared as a clean project/submission document.